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ME 5130: Advanced Heat and Mass Transfer
Homework 01: Heat Conduction
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1–2, Problem Statement: Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as:

$$\nabla T = \hat{\mathbf{u}}_\mathbf{r} \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin\theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} \tag{1}$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 q_r \right) + \frac{1}{r \sin\theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} \left(q_\theta \sin\theta \right) \tag{2}$$

1–2, Solution: The transformation from spherical to Cartesian coordinates is:

$$\begin{aligned} x &= r \sin\theta \cos\phi \\ y &= r \sin\theta \sin\phi \\ z &= r \cos\theta \end{aligned} \tag{3}$$

and the following can be defined:

$$\begin{aligned} u_1 &= r \\ u_2 &= \phi \\ u_3 &= \theta \end{aligned} \tag{4}$$

Now, from Özışık, we have Eq. (1–19):

$$\nabla T = \hat{\mathbf{u}}_\mathbf{1} \frac{1}{a_1} \frac{\partial T}{\partial u_1} + \hat{\mathbf{u}}_\mathbf{2} \frac{1}{a_2} \frac{\partial T}{\partial u_2} + \hat{\mathbf{u}}_\mathbf{3} \frac{1}{a_3} \frac{\partial T}{\partial u_3} \tag{5}$$

Eq. (1–22a):

$$\nabla \cdot \mathbf{q} = \frac{1}{a} \left[\frac{\partial}{\partial u_1} \left(\frac{a}{a_1} q_1 \right) + \frac{\partial}{\partial u_2} \left(\frac{a}{a_2} q_2 \right) + \frac{\partial}{\partial u_3} \left(\frac{a}{a_3} q_3 \right) \right] \tag{6}$$

Eq. (1–22b):

$$a = a_1 a_2 a_3 \tag{7}$$

and Equation (1–18):

$$a_i^2 = \left(\frac{\partial x}{\partial u_i} \right)^2 + \left(\frac{\partial y}{\partial u_i} \right)^2 + \left(\frac{\partial z}{\partial u_i} \right)^2, \qquad i = 1, 2, 3 \tag{8}$$

We can now compute the partial derivatives needed to find the scale factors:

$$\frac{\partial x}{\partial r} = \sin\theta \cos\phi \tag{9}$$

$$\frac{\partial y}{\partial r} = \sin\theta \sin\phi \tag{10}$$

$$\frac{\partial z}{\partial r} = \cos\theta \tag{11}$$

$$\frac{\partial x}{\partial \phi} = -r \sin\theta \sin\phi \tag{12}$$

$$\frac{\partial y}{\partial \phi} = r \sin\theta \cos\phi \tag{13}$$

$$\frac{\partial z}{\partial \phi} = 0 \tag{14}$$

$$\frac{\partial x}{\partial \theta} = r \cos\theta \cos\phi \tag{15}$$

$$\frac{\partial y}{\partial \theta} = r \cos\theta \sin\phi \tag{16}$$

$$\frac{\partial z}{\partial \theta} = -r \sin\theta \tag{17}$$

We can now evaluate:

$$\begin{aligned} a_r^2 &= \sin^2\theta \cos^2\phi + \sin^2\theta \sin^2\phi + \cos^2\theta \\ &= \sin^2\theta(\cos^2\phi + \sin^2\phi) + \cos^2\theta \\ &= \sin^2\theta + \cos^2\theta \\ &= 1 \\ a_1 = a_r &= 1 \end{aligned} \tag{18}$$

$$\begin{aligned} a_\phi^2 &= r^2 \sin^2\theta \sin^2\phi + r^2 \sin^2\theta \cos^2\phi \\ &= r^2 \sin^2\theta(\sin^2\phi + \cos^2\phi) \\ &= r^2 \sin^2\theta \\ a_2 = a_\phi &= r \sin\theta \end{aligned} \tag{19}$$

$$\begin{aligned} a_\theta^2 &= r^2 \cos^2\theta \cos^2\phi + r^2 \cos^2\theta \sin^2\phi + r^2 \sin^2\theta \\ &= r^2 \cos^2\theta(\cos^2\phi + \sin^2\phi) + r^2 \sin^2\theta \\ &= r^2 \cos^2\theta + r^2 \sin^2\theta \\ &= r^2(\cos^2\theta + \sin^2\theta) \\ &= r^2 \\ a_3 = a_\theta &= r \end{aligned} \tag{20}$$

We should note that since spherical coordinates (r, ϕ, θ) form an orthogonal coordinate system, we can use equations (1–19) and (1–22). Nevertheless, we can now rewrite equation (1–19):

$$\boxed{\nabla T = \hat{\mathbf{u}}_\mathbf{r} \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin\theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta}} \tag{21}$$

We can also now write:

$$a = a_1 a_2 a_3 = a_r a_\phi a_\theta = r^2 \sin\theta \tag{22}$$

Therefore:

$$\frac{a}{a_r} = r^2 \sin\theta \tag{23}$$

$$\frac{a}{a_\phi} = r \tag{24}$$

$$\frac{a}{a_\theta} = r \sin\theta \tag{25}$$

Therefore, equation (1–22a) can be rewritten as:

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2 \sin\theta} \left[\frac{\partial}{\partial r} \left(r^2 \sin\theta q_r \right) + \frac{\partial}{\partial \phi} \left(r q_\phi \right) + \frac{\partial}{\partial \theta} \left(r \sin\theta q_\theta \right) \right] \tag{26}$$

Which can be simplified to:

$$\boxed{\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 q_r \right) + \frac{1}{r \sin\theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta q_\theta \right)} \tag{27}$$

1–12, Problem Statement: Set up the mathematical formulation of the following heat conduction problems:

1–12, Solution:

1. A slab in $0 \leq x \leq L$ is initially at a temperature $F(x)$. For times $t > 0$, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ dissipates heat by convection into a medium at zero temperature.

   For a slab where $T(x, 0) = F(x)$ with no generation, the heat equation can be written as:

   $$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}, \qquad 0 < x < L,\ t > 0 \tag{28}$$

   At the $x = 0$ boundary condition:

   $$\frac{\partial T}{\partial x} = 0, \qquad x = 0,\ t > 0 \tag{29}$$

   At the $x = L$ boundary condition:

$$
\begin{aligned}
-k\frac{\partial T}{\partial x} &= h(T - T_\infty) \\
-k\frac{\partial T}{\partial x} &= hT \\
k\frac{\partial T}{\partial x} + hT &= 0, \qquad x = L,\, t > 0
\end{aligned} \tag{30}
$$

and finally, at the initial condition $t = 0$:

$$
T(x, 0) = F(x), \qquad 0 \leq x \leq L,\, t = 0 \tag{31}
$$

2. A semi-infinite region $0 \leq x < \infty$ is initially at a temperature $F(x)$. For times $t > 0$, heat is generated in the medium at a constant rate of $g_0 W/m^3$, while the boundary at $x = 0$ is kept at zero temperature.

   For a semi-infinite region with a constant rate of heat generation of $g_0 W/m^3$, the heat equation can be written as:

$$
\frac{\partial^2 T}{\partial x^2} + \frac{g_0}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad 0 < x < \infty,\, t > 0 \tag{32}
$$

   At the $x = 0$ boundary condition:

$$
T(x, t) = 0, \qquad x = 0,\, t > 0 \tag{33}
$$

   and finally, at the initial condition $t = 0$:

$$
T(x, 0) = F(x), \qquad 0 \leq x < \infty,\, t = 0 \tag{34}
$$

3. A solid cylinder $0 \leq r \leq b$ is initially at a temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3, while the boundary at $r = b$ dissipates heat by convection into a medium at zero temperature.

   For a solid cylinder with heat generated at a rate of $g(r)$, the heat equation can be written as:

$$
\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{g(r)}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad 0 \leq r < b,\, t > 0 \tag{35}
$$

   At the boundary condition $r = 0$, by symmetry:

$$
\frac{\partial T}{\partial r} = 0, \qquad r = 0,\, t > 0 \tag{36}
$$

   At the boundary condition $r = b$:

$$
\begin{aligned}
-k\frac{\partial T}{\partial r} &= h(T - T_\infty) \\
-k\frac{\partial T}{\partial r} &= hT \\
k\frac{\partial T}{\partial r} + hT &= 0, \qquad r = b,\, t > 0
\end{aligned}
\tag{37}
$$

and finally, at the initial condition $t = 0$:

$$
T(r, 0) = F(r), \qquad 0 \le r \le b,\, t = 0 \tag{38}
$$

4. A solid sphere $0 \le r \le b$ is initially at temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3, while the boundary at $r = b$ is kept at a uniform temperature T_0.

   For a solid sphere with heat generated at a rate of $g(r)$, the heat equation can be written as:

$$
\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right) + \frac{g(r)}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad 0 \le r < b,\, t > 0 \tag{39}
$$

   At the boundary condition $r = 0$, by symmetry:

$$
\frac{\partial T}{\partial r} = 0, \qquad r = 0,\, t > 0 \tag{40}
$$

   At the boundary condition $r = b$:

$$
T(b, t) = T_0, \qquad r = b,\, t > 0 \tag{41}
$$

   and finally, at the initial condition $t = 0$:

$$
T(r, 0) = F(r), \qquad 0 \le r \le b,\, t = 0 \tag{42}
$$

1–15, Problem Statement: A long cylindrical iron bar of diameter $D = 5\,\text{cm}$, initially at temperature $T_0 = 650\,^\circ\text{C}$, is exposed to an air stream at $T_\infty = 50\,^\circ\text{C}$. The heat transfer coefficient between the air stream and the surface of the bar is $h = 80\,\text{W}\,\text{m}^{-2}\,^\circ\text{C}^{-1}$. Thermophysical properties may be taken as $\rho = 7800\,\text{kg}\,\text{m}^{-3}$, $C_p = 460\,\text{J}\,\text{kg}^{-1}\,^\circ\text{C}^{-1}$, and $k = 60\,\text{W}\,\text{m}^{-1}\,^\circ\text{C}^{-1}$. Determine the time required for the temperature of the bar to reach $250\,^\circ\text{C}$ by using the lumped system analysis.

1–15, Solution:
First, we should check the Biot number to confirm the validity of the lumped analysis method:

$$
\text{Bi} = \frac{hL_c}{k} \tag{43}
$$

Where:

$$\begin{aligned} L_c &= \frac{V}{A_s} \\ &= \frac{\pi r^2 L}{2\pi r L} \\ &= \frac{r}{2} \\ &= \frac{0.025}{2} \\ &= 0.0125\text{m} \end{aligned} \tag{44}$$

We should note that since the bar is long compared to the diameter, we can ignore the surface area of the end faces in the calculation above. Therefore:

$$\begin{aligned} \text{Bi} &= \frac{hL_c}{k} \\ &= \frac{80 \cdot 0.0125}{60} \\ &\approx 0.0167 \end{aligned} \tag{45}$$

Since the Biot number is less than 0.1, the lumped system analysis is valid. Now, from Equation 1–69, we have:

$$\frac{\theta(t)}{\theta_0} = \frac{T(t) - T_\infty}{T_0 - T_\infty} = e^{-mt} \tag{46}$$

Where:

$$\begin{aligned} m &= \frac{hA_s}{\rho V C_p} \\ &= \frac{h}{\rho L_c C_p} \\ &= \frac{80}{7800 \cdot 0.0125 \cdot 460} \\ &\approx 1.784 \times 10^{-3}\,\text{s}^{-1} \end{aligned} \tag{47}$$

Therefore, we have:

$$\begin{aligned} \frac{250 - 50}{650 - 50} &= e^{-mt} \\ \frac{1}{3} &= e^{-mt} \\ -mt &= -\ln 3 \\ t &= \frac{\ln 3}{m} \\ t &= \frac{\ln 3}{1.784 \times 10^{-3}} \end{aligned} \tag{48}$$

Thus:

$$\boxed{t \approx 616\,\text{seconds}} \tag{49}$$

1–16, Problem Statement: A thermocouple is to be used to measure the temperature in a gas stream. The junction may be approximated as a sphere having thermal conductivity $k = 25\,\text{W}\,\text{m}^{-1}\,^\circ\text{C}^{-1}$, $\rho = 8400\,\text{kg}\,\text{m}^{-3}$, and $C_p = 0.4\,\text{kJ}\,\text{kg}^{-1}\,^\circ\text{C}^{-1}$. The heat transfer coefficient between the junction and the gas stream is $h = 560\,\text{W}\,\text{m}^{-2}\,^\circ\text{C}^{-1}$. Calculate the diameter of the junction if the thermocouple should measure 95% of the applied temperature difference in 3s.

1–16, Solution:
In order to measure 95% of the applied temperature difference, we must have:

$$\frac{T - T_\infty}{T_0 - T_\infty} = 0.05 \tag{50}$$

We can now use this to find:

$$0.05 = e^{-mt} \tag{51}$$

Where:

$$\begin{aligned} m &= \frac{hA_s}{\rho V C_p} \\ &= \frac{h}{\rho L_c C_p} \end{aligned} \tag{52}$$

and

$$\begin{aligned} L_c &= \frac{V}{A_s} \\ &= \frac{\frac{4}{3}\pi r^3}{4\pi r^2} \\ &= \frac{r}{3} \\ &= \frac{D}{6} \end{aligned} \tag{53}$$

Therefore, we can write:

$$\begin{aligned} m &= \frac{6h}{\rho D C_p} \\ &= \frac{6 \cdot 560}{8400 \cdot D \cdot 400} \\ &= \frac{0.001}{D} \end{aligned} \tag{54}$$

and further:

$$\begin{aligned} 0.05 &= e^{-3\left(\frac{0.001}{D}\right)} \\ \ln 0.05 &= -\frac{0.003}{D} \\ D &= \frac{-0.003}{\ln 0.05} \end{aligned} \tag{55}$$

and finally:

$$\boxed{D \approx 10^{-3}\,\mathrm{m} = 1\,\mathrm{mm}} \tag{56}$$

2–2, Problem Statement: A slab, $0 \leq x \leq L$, is initially at a temperature $F(x)$. For times $t > 0$ the boundary surface at $x = 0$ is kept insulated and that at $x = L$ dissipates heat by convection into a medium at zero temperature with a heat transfer coefficient h. Obtain an expression for the temperature distribution $T(x,t)$ in the slab for times $t > 0$ and for the heat flux at the boundary surface $x = L$.

2–2, Solution:
From problem 1-12, we know that for a slab where $T(x,0) = F(x)$ with no generation, the heat equation can be written as:

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad 0 < x < L,\, t > 0 \tag{57}$$

At the $x = 0$ boundary condition:

$$\frac{\partial T}{\partial x} = 0, \qquad x = 0,\, t > 0 \tag{58}$$

At the $x = L$ boundary condition, where $H = \frac{h}{k}$:

$$\frac{\partial T}{\partial x} + HT = 0, \qquad x = L,\, t > 0 \tag{59}$$

At the initial condition $t = 0$:

$$T(x,0) = F(x), \qquad 0 \leq x \leq L,\, t = 0 \tag{60}$$

Now, from Equation (2-36a) we know that the solution for the temperature $T(x,t)$ is:

$$T(x,t) = \sum_{m=1}^{\infty} e^{-\alpha\beta_m^2 t}\frac{1}{N(\beta_m)}X(\beta_m, x)\int_0^L X(\beta_m, x')\,F(x')dx' \tag{61}$$

At $x = 0$:

$$\begin{aligned} \frac{\partial T}{\partial x} &= 0 \\ X'(0)\Gamma(t) &= 0 \end{aligned} \tag{62}$$

and since $\Gamma(t) = e^{-\alpha\beta^2 t} \neq 0$:

$$\frac{dX}{dx} = 0 \tag{63}$$

Similarly, at $x = L$:

$$\begin{aligned} \frac{\partial T}{\partial x} + HT &= 0 \\ \frac{dX}{dx}\Gamma(t) + HX(L)\Gamma(t) &= 0 \\ \frac{dX}{dx} + HX(L) &= 0 \end{aligned} \tag{64}$$

Therefore from Table 2-2 No. 4 we know that:

$$X\left(\beta_m, x\right) = \cos \beta_m x \tag{65}$$

and:

$$\frac{1}{N\left(\beta_m\right)} = 2\frac{\beta_m^2 + H^2}{L\left(\beta_m^2 + H^2\right) + H} \tag{66}$$

where the eigenvalues β_m are the positive roots of:

$$\beta_m \tan \beta_m L = H \tag{67}$$

Therefore:

$$\boxed{T(x,t) = \sum_{m=1}^{\infty} e^{-\alpha\beta_m^2 t}\, 2\frac{\beta_m^2 + H^2}{L\left(\beta_m^2 + H^2\right) + H} \cos \beta_m x \int_0^L \cos \beta_m x' F(x') dx'} \tag{68}$$

Now, using Fourier's law:

$$q = -k\frac{\partial T}{\partial x} \tag{69}$$

The heat flux at the boundary surface $x = L$ can be expressed as:

$$\boxed{q = 2k \sum_{m=1}^{\infty} \beta_m e^{-\alpha\beta_m^2 t}\, \frac{\beta_m^2 + H^2}{L\left(\beta_m^2 + H^2\right) + H} \sin \beta_m L \int_0^L \cos \beta_m x' F(x') dx'} \tag{70}$$

2–6, Problem Statement: In a one-dimensional infinite medium, $-\infty < x < \infty$, initially, the region $a < x < b$ is at a constant temperature T_0, and everywhere outside this region is at zero temperature. Obtain an expression for the temperature distribution $T\left(x, t\right)$ in the medium for times $t > 0$.

2–6, Solution:
For a one-dimensional infinite medium, the heat equation can be written as:

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad -\infty < x < \infty,\, t > 0 \tag{71}$$

At $t = 0$:

$$\begin{aligned} T &= T_0, \qquad a < x < b \\ T &= 0, \qquad x \leq a \,\text{and}\, x \geq b \end{aligned} \tag{72}$$

From Equation 2–70 we have:

$$T(x,t) = \frac{1}{[4\pi\alpha t]^{1/2}} \int_{-\infty}^{\infty} F(x') \exp\left[-\frac{(x-x')^2}{4\alpha t}\right] dx' \tag{73}$$

Now, for the region where $x' \leq a$ and $x' \geq b$, the integrand above is zero, therefore:

$$T(x,t) = \frac{T_0}{[4\pi\alpha t]^{1/2}} \int_{a}^{b} \exp\left[-\frac{(x-x')^2}{4\alpha t}\right] dx' \tag{74}$$

Now, let:

$$\eta = \frac{x - x'}{\sqrt{4\alpha t}} \tag{75}$$

therefore:

$$dx' = -\sqrt{4\alpha t}\, d\eta \tag{76}$$

and

$$\begin{aligned} T(x,t) &= \frac{T_0}{[4\pi\alpha t]^{1/2}} \int_{(x-a)/\sqrt{4\alpha t}}^{(x-b)/\sqrt{4\alpha t}} e^{-\eta^2} \left(-\sqrt{4\alpha t}\right) d\eta \\ &= \frac{T_0}{\sqrt{\pi}} \int_{(x-b)/\sqrt{4\alpha t}}^{(x-a)/\sqrt{4\alpha t}} e^{-\eta^2}\, d\eta \end{aligned} \tag{77}$$

Which can be expressed as:

$$T(x,t) = \frac{T_0}{2}\left[\frac{2}{\sqrt{\pi}} \int_{0}^{(x-a)/\sqrt{4\alpha t}} e^{-\eta^2}\, d\eta - \frac{2}{\sqrt{\pi}} \int_{0}^{(x-b)/\sqrt{4\alpha t}} e^{-\eta^2}\, d\eta\right] \tag{78}$$

and since:

$$\mathbf{erf}(z) = \frac{2}{\sqrt{\pi}} \int_{0}^{z} e^{-\xi^2}\, d\xi \tag{79}$$

We have:

$$T(x,t) = \frac{T_0}{2}\left[\mathbf{erf}\left(\frac{x-a}{\sqrt{4\alpha t}}\right) - \mathbf{erf}\left(\frac{x-b}{\sqrt{4\alpha t}}\right)\right] \tag{80}$$

which can be rewritten as:

$$\boxed{\frac{T(x,t)}{T_0} = \frac{1}{2}\left[\mathbf{erf}\left(\frac{x-a}{\sqrt{4\alpha t}}\right) + \mathbf{erf}\left(\frac{b-x}{\sqrt{4\alpha t}}\right)\right], \qquad -\infty < x < \infty} \tag{81}$$

2–15, Problem Statement: Obtain an expression for the steady-state temperature distribution $T(x, y)$ in a rectangular region $0 \leq x \leq a$, $0 \leq y \leq b$ for the following boundary conditions: the boundary at $x = 0$ is kept insulated, the boundary at $y = 0$ is kept at temperature $f(x)$ and the boundaries at $x = a$ and $y = b$ dissipate heat by convection into an environment at zero temperature. Assume the heat transfer coefficient to be the same for both boundaries.

2–15, Solution:
For a steady-state two-dimensional rectangular problem, the heat equation can be written as:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0, \qquad 0 < x < a,\, 0 < y < b \tag{82}$$

At $x = 0$:

$$\frac{\partial T}{\partial x} = 0 \tag{83}$$

Let $H = \frac{h}{k}$. At $x = a$:

$$\frac{\partial T}{\partial x} + HT = 0 \tag{84}$$

At $y = 0$:

$$T(x, 0) = f(x) \tag{85}$$

At $y = b$:

$$\frac{\partial T}{\partial y} + HT = 0 \tag{86}$$

We can now use separation of variables to write:

$$T(x, y) = X(x)\, Y(y) \tag{87}$$

Then:

$$\begin{aligned} X''(x)Y + XY'' &= 0 \\ \frac{X''}{X} &= -\frac{Y''}{Y} \end{aligned} \tag{88}$$

Therefore:

$$\begin{aligned} \frac{d^2 X}{dx^2} + \beta^2 X &= 0 \\ \frac{d^2 Y}{dy^2} - \beta^2 Y &= 0 \end{aligned} \tag{89}$$

At $x = 0$:

$$\frac{dX}{dx} = 0 \tag{90}$$

At $x = a$:

$$\frac{dX}{dx} + HX(a) = 0 \tag{91}$$

At $y = 0$:

$$X(x)Y(0) = f(x) \tag{92}$$

At $y = b$:

$$\frac{dY}{dy} + HY(b) = 0 \tag{93}$$

For X, we can now use Table 2-2 No. 4 to find:

$$X(\beta_m, x) = \cos \beta_m x \tag{94}$$

$$\frac{1}{N(\beta_m)} = \frac{2(\beta_m^2 + H^2)}{a(\beta_m^2 + H^2) + H} \tag{95}$$

where the eigenvalues β_m are the positive roots of:

$$\beta_m \tan \beta_m a = H \tag{96}$$

For Y, we know the solution takes the form:

$$Y = C_3 \cosh \beta y + C_4 \sinh \beta y \tag{97}$$

Which can be rewritten as:

$$Y = C_3 \cosh \beta(b - y) + C_4 \sinh \beta(b - y) \tag{98}$$

We should note that this can be done because:

$$\begin{aligned} \frac{d}{dy} \cosh \beta(b - y) &= -\beta \sinh \beta(b - y) \\ \frac{d^2}{dy^2} \cosh \beta(b - y) &= \beta^2 \cosh \beta(b - y) \end{aligned} \tag{99}$$

and

$$\begin{aligned} \frac{d}{dy} \sinh \beta(b - y) &= -\beta \cosh \beta(b - y) \\ \frac{d^2}{dy^2} \sinh \beta(b - y) &= \beta^2 \sinh \beta(b - y) \end{aligned} \tag{100}$$

are both solutions of the differential equation. Now, at $y = b$:

$$\begin{aligned} Y &= C_3 \cosh \beta(0) + C_4 \sinh \beta(0) \\ &= C_3 \end{aligned} \tag{101}$$

$$\begin{aligned} \frac{dY}{dy} &= -\beta C_3 \sinh \beta(b-y) - \beta C_4 \cosh \beta(b-y) \\ &= -\beta C_3 \sinh \beta(0) - \beta C_4 \cosh \beta(0) \\ &= -\beta C_4 \end{aligned} \tag{102}$$

Therefore:

$$\begin{aligned} -\beta C_4 + H C_3 &= 0 \\ \beta C_4 &= H C_3 \\ C_4 &= \frac{H C_3}{\beta} \end{aligned} \tag{103}$$

and:

$$\begin{aligned} Y &= C_3 \cosh \beta(b-y) + \frac{H C_3}{\beta} \sinh \beta(b-y) \\ &= \frac{C_3}{\beta} \left[\beta \cosh \beta(b-y) + H \sinh \beta(b-y) \right] \end{aligned} \tag{104}$$

Since $\frac{C_3}{\beta}$ is constant (it will factor into the constant multiplier in the next steps), we can write:

$$Y(\beta_m, y) = \beta_m \cosh \beta_m(b-y) + H \sinh \beta_m(b-y) \tag{105}$$

We can now write the general solution as:

$$T(x,y) = \sum_{m=1}^{\infty} C_m \cos \beta_m x \left[\beta_m \cosh \beta_m(b-y) + H \sinh \beta_m(b-y) \right] \tag{106}$$

At $y = 0$:

$$f(x) = \sum_{m=1}^{\infty} C_m \cos \beta_m x \left[\beta_m \cosh \beta_m b + H \sinh \beta_m b \right], \qquad 0 < x < a \tag{107}$$

Let $K_m = \beta_m \cosh \beta_m b + H \sinh \beta_m b$:

$$f(x) = \sum_{m=1}^{\infty} C_m K_m \cos \beta_m x, \qquad 0 < x < a \tag{108}$$

We can then multiply both sides by the eigenfunction $\cos \beta_n x$:

$$\begin{aligned} f(x) \cos \beta_n x &= \sum_{m=1}^{\infty} C_m K_m \cos \beta_m x \cos \beta_n x \\ \int_0^a f(x) \cos \beta_n x \, dx &= \int_0^a \sum_{m=1}^{\infty} C_m K_m \cos \beta_m x \cos \beta_n x \, dx \end{aligned} \tag{109}$$

We can now recall:

$$\int_0^a X_m(\beta_m, x) X_n(\beta_n, x) dx = \begin{cases} 0 & \text{if } n \neq m \\ N(\beta_m) & \text{if } n = m \end{cases} \tag{110}$$

Therefore:

$$\begin{aligned} \int_0^a f(x) \cos \beta_n x \, dx &= C_m K_m N(\beta_m) \\ C_m &= \frac{1}{K_m N(\beta_m)} \int_0^a f(x) \cos \beta_n x \, dx \\ &= \frac{1}{N(\beta_m)[\beta_m \cosh \beta_m b + H \sinh \beta_m b]} \int_0^a f(x') \cos \beta_m x' \, dx' \end{aligned} \tag{111}$$

Finally, we have:

$$\boxed{T(x,y) = \sum_{m=1}^{\infty} \frac{\cos \beta_m x \left[\beta_m \cosh \beta_m (b-y) + H \sinh \beta_m (b-y)\right]}{N(\beta_m)[\beta_m \cosh \beta_m b + H \sinh \beta_m b]} \int_0^a \cos \beta_m x' f(x') \, dx'} \tag{112}$$

Where:

$$\frac{1}{N\left(\beta_m\right)} = \frac{2\left(\beta_m^2 + H^2\right)}{a\left(\beta_m^2 + H^2\right) + H} \tag{113}$$

and the eigenvalues β_m are the positive roots of:

$$\beta_m \tan \beta_m a = H \tag{114}$$